9/10/25

Conditional Expectation

"Random variable"

Ex:

y \ x	-1	0	1
-1	1/2	1/4	1/8
1	1/16	1/16	0

$Y|X=-1 \sim \{-1, 1\}$ with probs $\{8/9, 1/9\}$

$Y|X=0 \sim \{-1, 1\}$ with probs $\{4/5, 1/5\}$

$Y|X=1 \sim \{-1\}$ with prob $\{1\}$

$E[Y|X=-1] = -7/9$

$E[Y|X=0] = -3/5$

$E[Y|X=1] = -1$

x	$g(x)$
-1	$-7/9$
0	$-3/5$
1	-1

$E[Y|X] \sim \{-7/9, -3/5, -1\}$ with probs $\{9/16, 1/16, 1/8\}$

General

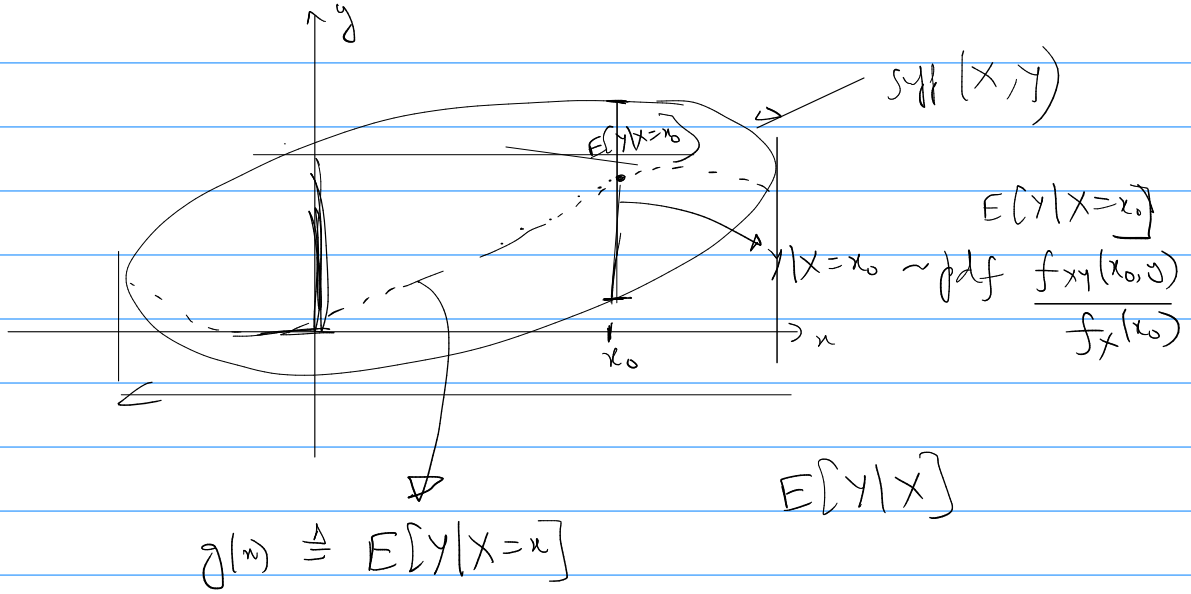
$Y|X=x$: distribution

$g(x) \triangleq E[Y|X=x]$: function of x

say $g(x)$

Conditional expectation of Y given X

$E[Y|X] \triangleq g(X)$



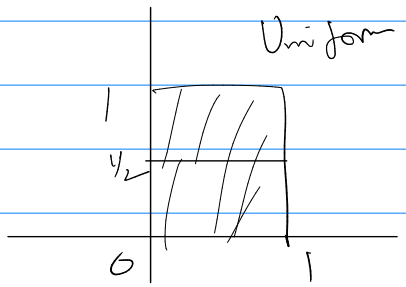
$Y|X=x_0 \sim$ pdf $\frac{f_{XY}(x_0, y)}{f_X(x_0)}$

$E[Y|X=x_0]$

$E[Y|X]$

$g(x) \triangleq E[Y|X=x]$

Ex: (1)



$E[Y|X=x_0] = 1/2$

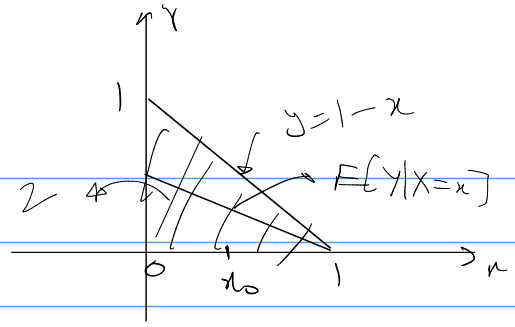
$x_0 \in [0,1]$

replace x_0 with X

$E[Y|X] = 1/2$ w.p. 1

$\sim \{1/2\}$ with prob $\{1\}$

(2)



$E[Y|X=x_0] = \frac{1-x_0}{2}$

$X \sim pdf\ 2(1-x)$

$x \in [0,1]$

$\boxed{E[Y|X] = \frac{1-X}{2}}$

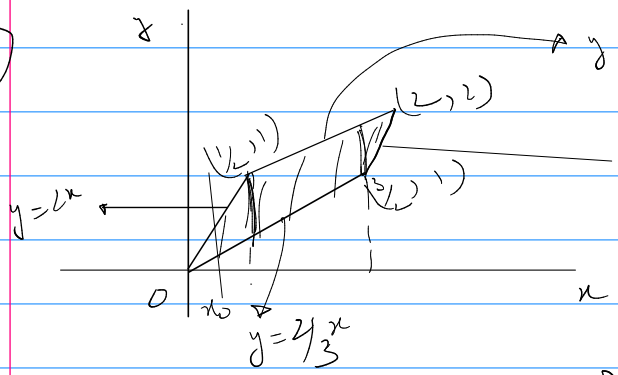
$\in [0, 1/2]$

$X \sim pdf\ 2(1-x) \quad x \in [0,1]$

$Y = g(X) = \frac{1-X}{2} \quad \in [0, 1/2]$

$f_Y(y) = \frac{f_X(1-2y)}{|-1/2|} = 8y, \quad y \in [0, 1/2]$

(3)



$$E[Y|X=x_0] = \begin{cases} 4x_0/3 & x_0 \in (0, 1/2) \\ \frac{2x_0}{3} + 1/3 & x_0 \in [1/2, 3/2] \\ ? & x_0 \in (3/2, 2) \end{cases}$$

Properties:

(1) "Tower" $\quad E[E[Y|X]] = E[Y]$

Ex: $X_1, X_2, \ldots \sim$ iid Bernulli(p)

$Y = \min\{k : X_k = 1\} \sim$ Geometric(p)

$$E[Y|X_1=1]=1 \qquad E[Y|X_1=0]=1+E[Y]$$

$$E[Y|X_1] \sim \begin{cases} p & 1-p \\ 1 & 1+E[Y] \end{cases}$$

$$E[Y]=E\big[E[Y|X_1]\big]=E[Y|X_1=0]\,P(X_1=0)+E[Y|X_1=1]\,P(X_1=1)$$

$$1+E[Y] \qquad (1-p) \qquad 1 \qquad p$$

$$E[Y]=1/p$$

$p=1/2$

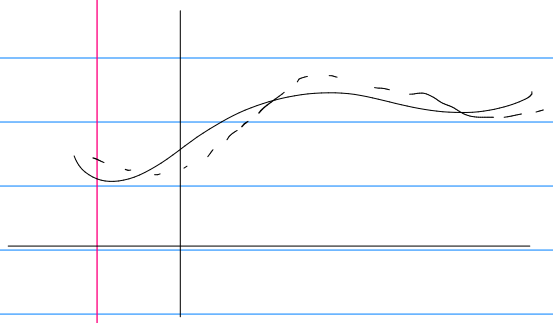
$$\longrightarrow \quad Z=\min\{k : X_{k-1}=0, X_k=1\}$$

$$E[Z]=?$$

(2) Prediction X, Y: joint distribution

Observe X. Want to predict Y using only X & the joint distribution.

Looking for ~ function of X s.t. $g(X)$ "predicts" Y well.
(say $g(X)$)



Mean square error (MSE)
squared error

$$E\big[(Y-g(X))^2\big]$$
error

MSE

$$E[Y|X]=\arg\min_g E\big[(Y-g(X))^2\big]$$

